Exercise 4: Worked solutions

Mathematics for Engineers I · Module 62765 · Exercise Sheet 4

From Fundamentals to AI Apocalypse · personal study guide

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Exercise 4.1: Mathematical induction

Induction starts from the first valid index and shows that each true case forces the next.

Connected topic: Induction (§2.5).

1a: Prove the finite geometric sum

Try first: check n=0, then multiply the proposed sum by x and subtract, keeping x≠1 visible.

Answers: see "Exercise 4.1: Mathematical induction" under Answers at the end.

1b: Prove the reciprocal-square-root bound

Try first: start the induction at $n = 2$, then show that the new term $1/\sqrt{n+1}$ is at least the step from $\sqrt{n}$ to $\sqrt{n+1}$.

Answers: see "Exercise 4.1: Mathematical induction" under Answers at the end.

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Exercise 4.2: Maps

Composition applies f first and g second. For finite-set counts, choose an image for each input while respecting the required property.

Connected topic: Map types and injectivity (§3.2) and counting and binomial coefficients (§2.4).

2a: Is a composition of injections injective?

Try first: assume two composite outputs are equal and unwind the definition of injectivity in reverse order.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

2b(i): Count all maps $M \to N$

Try first: each of the m elements of M independently chooses one of n target elements.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

2b(ii): Count bijections $M \to N$

Try first: A bijection requires equal cardinalities; when they match, count the possible orderings.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

2b(iii): Count injective maps $M \to N$

Try first: an injection assigns distinct targets to inputs, so there are fewer choices at each step.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

2b(iv): Count surjective maps $M \to N$

Try first: exclude maps whose image misses at least one target, correcting for maps that miss several targets.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

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Exercise 4.3: Elementary functions

Addition formulas for sine and cosine imply the circle identity and double-angle relation.

Connected topic: Elementary functions and inverse functions (§3.5).

3a: Prove the circle equation

Try first: expand $\cos(x-x)$ using the cosine addition theorem.

Answers: see "Exercise 4.3: Elementary functions" under Answers at the end.

3b: Prove the cosine double-angle identity

Try first: use $\cos(2x)=\cos(x+x)$, then replace $\cos^2 x$ by $1-\sin^2 x$.

Answers: see "Exercise 4.3: Elementary functions" under Answers at the end.

3c: Evaluate cos(arcsin *x*)

Try first: let $\theta=\arcsin x$, so $\sin\theta=x$ and θ lies in the principal interval where cosine is non-negative.

Answers: see "Exercise 4.3: Elementary functions" under Answers at the end.

3d: Find cos(π/5) exactly

Try first: use $\pi/5=2(\pi/10)$ in the double-angle identity, then select the positive root because $\pi/5$ is in the first quadrant.

Answers: see "Exercise 4.3: Elementary functions" under Answers at the end.

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Self-Study 4.4: Maps

Vertex form reveals the range and the restriction needed for a quadratic to become one-to-one.

Connected topic: Injectivity and inverses (§3.2), composition and domains (§3.4) and elementary functions (§3.5).

4a: Classify *f*, *g*, and *h*; modify *f* and *g*

Try first: complete the square for f, solve g for its input, and inspect the coordinate relation defining h.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

4b(i): Form $f \circ \mathrm{e}$

Try first: since $\mathrm{e}(x)=\mathrm{e}^x$ is always positive, substitute it into f and use the range of the downward parabola.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

4b(ii): Form $g \circ f$

Try first: the inner output must satisfy $f(x)\geq 0$ because g accepts only non-negative inputs.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

4b(iii): Form $g \circ \mathrm{e}$

Try first: check the exponential output against g's non-negative domain.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

4b(iv): Form $g \circ g$

Try first: the first g maps $[0,\infty)$ into $(0,1]$, which is contained in the domain $[0,\infty)$ of the second g.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

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Self-Study 4.5: Maps

Use the definitions directly: equality of composite values, existence of preimages, and the order of quantifiers.

Connected topic: Injectivity and surjectivity (§3.2).

5i: If $g \circ f$ is injective, show f is injective

Try first: assume f sends two inputs to the same value, then apply g.

Answers: see "Self-Study 4.5: Maps" under Answers at the end.

5ii: If $g \circ f$ is surjective, show g is surjective

Try first: take an arbitrary element of Z and use surjectivity of the composite to produce a preimage.

Answers: see "Self-Study 4.5: Maps" under Answers at the end.

5iii: If f and g are surjective, show $g \circ f$ is surjective

Try first: for each target z, first choose a g-preimage y, then choose an f-preimage x of y.

Answers: see "Self-Study 4.5: Maps" under Answers at the end.

5iv: Give an injective composite with non-injective g

Try first: make the domain of f a singleton, while g identifies two different middle elements.

Answers: see "Self-Study 4.5: Maps" under Answers at the end.

Self-Study 4.6: Elementary functions

Factor the quartic as a quadratic in x^2. For the parameterized quadratic, use the discriminant and include degenerate cases.

Connected topic: Polynomials and elementary functions (§3.5).

6a: Find and factor the roots

Try first: set $u{=}x^2$ for the quartic, then split the parameter cases for g_a.

Answers: see "Self-Study 4.6: Elementary functions" under Answers at the end.

6b: Show $\cos(3x)$ is not generally $3\cos x$

Try first: A single convenient counterexample disproves a universal equality; then derive the general expression.

Answers: see "Self-Study 4.6: Elementary functions" under Answers at the end.

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Self-Study 4.7: Strictly monotone functions

Strict monotonicity forbids equal outputs at distinct inputs; injectivity alone does not require order preservation.

Connected topic: Injectivity and inverse functions (§3.2).

7a: Increasing implies injective

Try first: for two distinct inputs, order them and apply the strict inequality.

Answers: see "Self-Study 4.7: Strictly monotone functions" under Answers at the end.

7b: Decreasing implies injective

Try first: repeat the order argument with inequalities reversed.

Answers: see "Self-Study 4.7: Strictly monotone functions" under Answers at the end.

7c: Is every injection strictly monotone?

Try first: find an injective rearrangement of the real line that reverses order for one pair.

Answers: see "Self-Study 4.7: Strictly monotone functions" under Answers at the end.

7d: Conclude that exp has an inverse

Try first: combine strict increase with the known range of the exponential.

Answers: see "Self-Study 4.7: Strictly monotone functions" under Answers at the end.

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Self-Study 4.8: Tangent addition theorems

Divide the sine and cosine addition identities only when the stated cosines are nonzero.

Connected topic: Trigonometric functions (§3.5).

8a: Prove the tangent addition and subtraction formulas

Try first: write $\tan(x\pm y)=\sin(x\pm y)/\cos(x\pm y)$, then divide numerator and denominator by $\cos x \cos y$.

Answers: see "Self-Study 4.8: Tangent addition theorems" under Answers at the end.

8b: Derive the tangent double-angle formula

Try first: set $y=x$ in the addition formula, observing the stated conditions keep the tangent defined.

Answers: see "Self-Study 4.8: Tangent addition theorems" under Answers at the end.

8c: Compute tan(π/12) exactly

Try first: apply the subtraction formula to $\pi/3-\pi/4$ and simplify by rationalizing.

Answers: see "Self-Study 4.8: Tangent addition theorems" under Answers at the end.

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Challenge 4.A: Surjective maps

Let $s(k)$ count maps onto a fixed k-element set. Partition all maps according to their image, then invert the resulting triangular relation.

Connected topics: Map types (§3.2), binomial coefficients and finite sums (§2.4) and induction (§2.5).

4.A setup: Interpret $s(k)$ and the target

Try first: relate surjectivity to the size of the image, and identify $s(n)$ with the desired count.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 1: Derive the recurrence for $s(k)$

Try first: partition the k^m maps by their exact image size j and solve for the $j{=}k$ term.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 2: Check the proposed formula for $k{=}1$, 2, and 3

Try first: use the recurrence to calculate the first values and compare them with the binomial sum.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 3a: Reverse the order of a triangular sum

Try first: list the index region rather than manipulating terms.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 3b: Prove the binomial coefficient identity

Try first: write both sides using factorial definitions.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 3c: Establish the convolution identity

Try first: after changing order, the coefficient of each i contains a complete alternating binomial sum.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 4: Prove $s(k){=}g(k)$ by induction

Try first: use $k{=}1$ as the base, then subtract the same lower-index terms from the recurrence and convolution identity.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

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Answers

Exercise 4.1: Mathematical induction

1a · Prove the finite geometric sum

1. Let $S_n=\sum_{k=0}^{n}x^k$. The claim is $S_n=(1-x^{n+1})/(1-x)$, for $x\neq1$.
2. At $n=0$, $S_0=1$ and $(1-x)/(1-x)=1$.
3. Assume the formula for n. Add the next term: $S_{n+1}=S_n+x^{n+1}=(1-x^{n+1})/(1-x)+x^{n+1}$.
4. Put the terms over the same denominator: $[1-x^{n+1}+x^{n+1}(1-x)]/(1-x)=(1-x^{n+2})/(1-x)$.
5. That is the claimed formula with $n+1$. The denominator is nonzero because $x\neq1$; the base and step prove the result for all $n\geq0$.

$$\sum_{k=0}^{n} x^k = \frac{1 - x^{n+1}}{1 - x}$$

Exercise 4.1: Mathematical induction

1b · Prove the reciprocal-square-root bound

1. Write S_n for the sum. The claim starts at $n = 2$, so that is the base case: $S_2 = 1 + 1/\sqrt{2} \approx 1.707$, while $\sqrt{2} \approx 1.414$. The claim holds at $n = 2$.

$$S_n = \sum_{k=1}^{n} \frac{1}{\sqrt{k}}$$

2. **Induction hypothesis:** assume $S_n > \sqrt{n}$ for some $n \geq 2$. Adding the next term gives

$$S_{n+1} = S_n + \frac{1}{\sqrt{n+1}} > \sqrt{n} + \frac{1}{\sqrt{n+1}}$$

3. It is enough to show that the right-hand side is at least $\sqrt{n+1}$. Multiply by the positive number $\sqrt{n+1}$:

$$\sqrt{n(n+1)} + 1 \geq n + 1 \Leftrightarrow \sqrt{n(n+1)} \geq n$$

   This holds because $n(n+1) > n^2$.

4. So $S_{n+1} > \sqrt{n+1}$: the claim at n passes to $n+1$, and with the base case it holds for every $n \geq 2$.

5. **Check:** at $n = 3$ the sum is $1 + 0.707 + 0.577 \approx 2.28$, above $\sqrt{3} \approx 1.73$. At $n = 1$ both sides equal 1, which is why the strict inequality starts at 2.

Exercise 4.2: Maps

2a · Is a composition of injections injective?

1. Suppose $(g\circ f)(x_1)=(g\circ f)(x_2)$. This says $g(f(x_1))=g(f(x_2))$.
2. Since g is injective, $f(x_1)=f(x_2)$. Since f is injective, $x_1=x_2$.
3. Thus $g\circ f$ is injective. The codomains need not be equal: the key is that each map is injective on the outputs that occur.

$$(g \circ f)(x_1) = (g \circ f)(x_2) \Rightarrow x_1 = x_2$$

Exercise 4.2: Maps

2b(i) · Count all maps $M\to N$

1. For each input $x\in M$, a map may assign any of the n elements of N.
2. There are n choices for each of m independent inputs, so the product rule gives $n\cdot n\cdots n=n^m$ maps.
3. If $n=0$ and $m>0$, this gives 0 maps, as expected; for $m=0$ the unique empty map is counted by $0^0=1$ under the combinatorial convention.

$$\#(M \to N) = n^m$$

Exercise 4.2: Maps

2b(ii) · Count bijections $M\to N$

1. If $m\neq n$, no bijection exists, so the count is 0.
2. If $m=n$, list the m inputs in any fixed order. The first image has m choices, the next $m-1$, and so on to 1.
3. The product is $m!$, including $0!=1$ for the empty sets.

$$\#\{\text{bijective maps } M \to N\} = m! \text{ when } m = n, \text{ otherwise } 0$$

Exercise 4.2: Maps

2b(iii) · Count injective maps $M \to N$

1. If $m>n$, the pigeonhole principle makes an injection impossible; the count is 0.
2. For $m \le n$, order the inputs. Choose distinct images in sequence: n choices, then $n-1$, down to $n-m+1$.
3. Therefore the count is $n(n-1)\dots(n-m+1) = n!/(n-m)!$. For $m=0$ this is 1.

$$\#\{\text{injective maps } M \to N\} = \frac{n!}{(n-m)!} \text{ when } m \le n, \text{ otherwise } 0$$

Exercise 4.2: Maps

2b(iv) · Count surjective maps $M \to N$

1. Start with all n^m maps. For a chosen set of j target elements forbidden from the image, the remaining $n-j$ targets give $(n-j)^m$ maps.
2. There are $\mathrm{C}(n,j)$ ways to choose those j omitted targets. Inclusion–exclusion alternates the correction by the number omitted.
3. The number of onto maps is $\sum_{j=0}^{n} (-1)^j \, \mathrm{C}(n,j)(n-j)^m$. This includes the empty-set cases when interpreted with the usual combinatorial convention.

$$\#\{\text{onto maps } M \to N\} = \sum_{j=0}^{n} (-1)^j \, \mathrm{C}(n,\, j)(n-\, j)^m$$

Exercise 4.3: Elementary functions

3a · Prove the circle equation

1. The cosine difference formula is $\cos(a-b) = \cos a \cos b + \sin a \sin b$.
2. Set $a=b=x$. The left side is $\cos 0 = 1$, and the right side is $\cos^2 x + \sin^2 x$.
3. Hence $\cos^2 x + \sin^2 x = 1$ for every real x.

$$\cos^2 x + \sin^2 x = 1$$

Exercise 4.3: Elementary functions

3b · Prove the cosine double-angle identity

1. The addition formula gives $\cos(x+x)=\cos^2 x-\sin^2 x$.
2. Using $\cos^2 x=1-\sin^2 x$ from the circle equation, $\cos(2x)=1-\sin^2 x-\sin^2 x=1-2\sin^2 x$.

$$\cos(2x) \;=\; 1 \;-\; 2\sin^2 x$$

Exercise 4.3: Elementary functions

3c · Evaluate cos(arcsin *x*)

1. By definition, $\theta=\arcsin x$ means $\sin\theta=x$, with $\theta\in[-\pi/2,\pi/2]$.
2. The circle identity gives $\cos^2\theta=1-\sin^2\theta=1-x^2$. On this interval $\cos\theta\geq 0$, so take the non-negative square root.
3. Therefore $\cos(\arcsin x)=\sqrt{1-x^2}$, valid for $-1\leq x\leq 1$.

$$\cos(\arcsin x) \;=\; \sqrt{1-x^2}, \text{ for } -1 \;\leq\; x \;\leq\; 1$$

Exercise 4.3: Elementary functions

3d · Find cos(π/5) exactly

1. The given value is $\sin(\pi/10)=(\sqrt{5}-1)/4$. Its square is $(3-\sqrt{5})/8$.
2. Apply $\cos(2x)=1-2\sin^2 x$ with $x=\pi/10$: $\cos(\pi/5)=1-2(3-\sqrt{5})/8=(1+\sqrt{5})/4$.
3. The result is positive, as expected for $\pi/5\in(0,\pi/2)$.

$$\cos(\pi/5) \;=\; \frac{1+\sqrt{5}}{4}$$

Self-Study 4.4: Maps

4a · Classify *f*, *g*, and *h*; modify *f* and *g*

1. For $f(x)=-x^2+6x-5$, complete the square: $f(x)=-(x-3)^2+4$. Its range is $(-\infty,4]$; it is not injective on $\mathbb{R}$ because points equally spaced about 3 have equal values.
2. Restrict f to $[3,\infty)$ and use codomain $(-\infty,4]$. It is then strictly decreasing and onto that codomain. Solving $y=4-(x-3)^2$ on this branch gives $f^{-1}(y)=3+\sqrt{4-y}$, $y\leq 4$.
3. For $g(x)=1/(1+x)$, $x\geq 0$, the values are $(0,1]$. It is strictly decreasing and injective, but not onto $\mathbb{R}$. With codomain $(0,1]$, it is bijective; solving $y=1/(1+x)$ gives $g^{-1}(y)=1/y-1$.
4. For $h(x,y)=(x,x+y,y)$, equality of outputs gives equality of the first and third coordinates, hence x and y agree: h is injective. Its image is the plane $\{(a,b,c):b=a+c\}$, so it is not onto $\mathbb{R}^3$ and is not bijective.

$$f(x) = -(x-3)^2 + 4$$

$$f^{-1}(y) = 3 + \sqrt{4-y}$$

5.

$$g^{-1}(y) = \frac{1}{y} - 1$$

Self-Study 4.4: Maps

4b(i) · Form *f*∘e

1. Every real x is allowed: $e^x\in\mathbb{R}$, the domain of f.
2. $(f\circ e)(x)=-(e^x)^2+6e^x-5=-e^{2x}+6e^x-5$.
3. Writing $t=e^x>0$ gives $4-(t-3)^2$. Its maximum 4 occurs when $t=3$; as $t\to 0^+$ or $t\to\infty$ the value tends without bound downward. The range is $(-\infty,4]$.

$$(f\circ \mathrm{e})(x) = -\mathrm{e}^{2x} + 6\mathrm{e}^x - 5, \text{ range } (-\infty, 4]$$

Self-Study 4.4: Maps

4b(ii) · Form *g*∘*f*

1. Require $4-(x-3)^2\geq 0$, so $(x-3)^2\leq 4$ and $1\leq x\leq 5$. This is the composite domain.
2. On it, $f(x)$ ranges over $[0,4]$, and $(g\circ f)(x)=1/(1+f(x))=1/[5-(x-3)^2]$.
3. The denominator runs from 1 to 5; reciprocation reverses the order. The range is $[1/5,1]$.

$$(g\circ f)(x) = \frac{1}{5-(x-3)^2}, \text{ domain } [1,\,5], \text{ range } \left[\frac{1}{5},\,1\right]$$

Self-Study 4.4: Maps

4b(iii) · Form $g \circ e$

1. For all $x \in \mathbb{R}$, $e^x > 0$, so the composition is defined everywhere.
2. $(g \circ e)(x) = 1/(1+e^x)$.
3. As e^x ranges over $(0,\infty)$, this expression decreases through $(0,1)$. It approaches 1 and 0 but reaches neither, so the range is $(0,1)$.

$$(g \circ \mathrm{e})(x) = \frac{1}{1 + \mathrm{e}^x}, \text{ range } (0,\ 1)$$

Self-Study 4.4: Maps

4b(iv) · Form $g \circ g$

1. The composite domain is $[0,\infty)$, since $g(x) \geq 0$ for every allowed x.
2. $(g \circ g)(x) = 1/[1+1/(1+x)] = (1+x)/(2+x)$.
3. This increases from $1/2$ at $x=0$ and approaches 1 without reaching it. The range is $[1/2,1)$.

$$(g \circ g)(x) = \frac{1 + x}{2 + x}, \text{ range } [\frac{1}{2},\ 1)$$

Self-Study 4.5: Maps

5i · If $g \circ f$ is injective, show f is injective

1. Suppose $f(x_1) = f(x_2)$. Applying g gives $(g \circ f)(x_1) = (g \circ f)(x_2)$.
2. Injectivity of $g \circ f$ implies $x_1 = x_2$. Thus f is injective.

$$g \circ f \text{ injective} \Rightarrow f \text{ injective}$$

Self-Study 4.5: Maps

5ii · If $g \circ f$ is surjective, show g is surjective

1. Let $z \in Z$. Since $g \circ f$ is onto Z, some $x \in X$ satisfies $g(f(x)) = z$.
2. The element $y = f(x)$ belongs to Y and $g(y) = z$. Every z has a preimage under g, so g is surjective.

$$g \circ f \text{ surjective} \Rightarrow g \text{ surjective}$$

Self-Study 4.5: Maps

5iii · If f and g are surjective, show $g \circ f$ is surjective

1. Let $z \in Z$. Surjectivity of g gives $y \in Y$ with $g(y)=z$.
2. Surjectivity of f gives $x \in X$ with $f(x)=y$. Therefore $g(f(x))=z$.
3. Every z is reached by $g \circ f$, proving surjectivity.

$$f \text{ and } g \text{ surjective} \Rightarrow g \circ f \text{ surjective}$$

Self-Study 4.5: Maps

5iv · Give an injective composite with non-injective g

1. Let $X=\{a\}$, $Y=\{0,1\}$, $Z=\{z\}$. Define $f(a)=0$ and define $g(0)=g(1)=z$.
2. The map g is not injective because $0 \neq 1$ but $g(0)=g(1)$.
3. The composite $X \to Z$ sends the only input a to z. A map on a singleton is injective, so $g \circ f$ is injective.

$$X = \{a\}, Y = \{0, 1\}, Z = \{z\}$$

$$f(a) = 0, g(0) = g(1) = z$$

Made by Thanate A

Self-Study 4.6: Elementary functions

6a · Find and factor the roots

1. For f, divide by 5 and set $u = x^2$: $u^2 + 4u - 5 = (u - 1)(u + 5)$. Thus $f(x) = 5(x^2 - 1)(x^2 + 5)$.
2. The factor $x^2 - 1$ gives the real roots $x = \pm 1$; $x^2 + 5$ is always positive, and over $\mathbb{C}$ it has the roots $\pm i\sqrt{5}$. The complete linear factorisation is

$$f(x) = 5(x - 1)(x + 1)(x - i\sqrt{5})(x + i\sqrt{5})$$

3. For $g_a(x) = ax^2 + 2x + 1$: if $a = 0$ it is the line $2x + 1$, with the single root $-1/2$. If $a \neq 0$, the discriminant decides:

$$\Delta = 2^2 - 4a = 4(1 - a)$$

4. When $a < 1$ and $a \neq 0$, $\Delta > 0$ and there are two distinct real roots, so $g_a = a(x - r_+)(x - r_-)$ with

$$r_\pm = \frac{-1 \pm \sqrt{1 - a}}{a}$$

   At $a = 1$, $\Delta = 0$ and $g_a = (x + 1)^2$, a double root at -1.
5. When $a > 1$, $\Delta < 0$: there are no real roots, and over $\mathbb{C}$ the same factorisation holds with $r_\pm = (-1 \pm i\sqrt{a - 1})/a$.
6. **Number of real roots of g_a:** two for $a < 1$ with $a \neq 0$; one for $a = 0$ (a line) and for $a = 1$ (a double root); none for $a > 1$.

Self-Study 4.6: Elementary functions

6b · Show cos(3x) is not generally 3cos x

1. Take x=0: cos(3·0)=1 while 3cos 0=3, so they are not equal in general.
2. Use cos(2x+x)=cos(2x)cos x−sin(2x)sin x. Substitute cos2x=cos^2x−sin^2x and sin2x=2sin x cos x.
3. This gives cos3x=cos^3x−3sin^2x cos x=cos^3x−3(1−cos^2x)cos x=4cos^3x−3cos x.

$$\cos(3x) = 4\cos^3 x - 3\cos x$$

Self-Study 4.7: Strictly monotone functions

7a · Increasing implies injective

1. Take $x \neq y$. Either $x<y$ or $y<x$. If $x<y$, strict increase gives $f(x)<f(y)$, hence $f(x)\neq f(y)$.
2. If $y<x$, strict increase gives $f(y)<f(x)$, again unequal outputs. Therefore f is injective.

$$x < y \Rightarrow f(x) < f(y) \Rightarrow f(x) \neq f(y)$$

Self-Study 4.7: Strictly monotone functions

7b · Decreasing implies injective

1. Take $x \neq y$. If $x<y$, strict decrease gives $f(x)>f(y)$; if $y<x$, it gives $f(y)>f(x)$.
2. In either case the outputs differ, so f is injective.

$$x < y \Rightarrow f(x) > f(y) \Rightarrow f(x) \neq f(y)$$

Self-Study 4.7: Strictly monotone functions

7c · Is every injection strictly monotone?

1. Define $f:\mathbb{R}\to\mathbb{R}$ by swapping 0 and 1 and leaving every other real number fixed. This is a bijection, hence injective.
2. Since $0<1/2$ but $f(0)=1>f(1/2)=1/2$, it is not increasing. Since $1/2<1$ but $f(1/2)=1/2>f(1)=0$, it is not decreasing.
3. Thus injectivity does not imply strict monotonicity.

$$f(0) = 1 > 1/2 = f(1/2)$$

$$f(1/2) = 1/2 > 0 = f(1)$$

Self-Study 4.7: Strictly monotone functions

7d · Conclude that exp has an inverse

1. The exponential is strictly increasing on $\mathbb{R}$, so by part (a) it is injective.
2. Its range is $(0,\infty)$, so $\exp:\mathbb{R}\to(0,\infty)$ is also surjective.
3. It is therefore bijective and has an inverse, the natural logarithm $\ln:(0,\infty)\to\mathbb{R}$.

$$\exp : \mathbb{R} \to (0,\, \infty) \text{ is bijective, } \exp^{-1} = \ln$$

Self-Study 4.8: Tangent addition theorems

8a · Prove the tangent addition and subtraction formulas

1. The addition identities give $\sin(x\pm y)=\sin x\cos y\pm\cos x\sin y$ and $\cos(x\pm y)=\cos x\cos y\mp\sin x\sin y$.
2. The hypotheses ensure $\cos x$, $\cos y$, and $\cos(x\pm y)$ are nonzero, so division is valid. Divide top and bottom by $\cos x\cos y$.
3. The result is $\tan(x\pm y)=(\tan x\pm\tan y)/(1\mp\tan x\tan y)$. For the plus case use matching upper/lower signs; for the minus case use opposite signs.

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

Self-Study 4.8: Tangent addition theorems

8b · Derive the tangent double-angle formula

1. Use $\tan(x+x)=(\tan x+\tan x)/(1-\tan^2 x)$.
2. Thus $\tan(2x)=2\tan x/(1-\tan^2 x)$, whenever x and $2x$ avoid the excluded tangent poles.

$$\tan(2x) = \frac{2\tan x}{1 - \tan^2 x}$$

Self-Study 4.8: Tangent addition theorems

8c · Compute tan(π/12) exactly

1. Use $\tan(\pi/3)=\sqrt{3}$ and $\tan(\pi/4)=1$. Then $\tan(\pi/12)=(\sqrt{3}-1)/(1+\sqrt{3})$.
2. Multiply numerator and denominator by $\sqrt{3}-1$: the denominator becomes $3-1=2$ and the numerator becomes $(\sqrt{3}-1)^2=4-2\sqrt{3}$.
3. Therefore $\tan(\pi/12)=2-\sqrt{3}$, a positive value as expected for an angle of 15°.

$$\tan(\pi/12) = 2 - \sqrt{3}$$

Challenge 4.A: Surjective maps

4.A setup · Interpret $s(k)$ and the target

1. For a fixed k-element set A, $s(k)$ is the number of maps $f{:}M{\to}A$ whose image is all of A.
2. A map $M{\to}N$ is surjective exactly when its image has n elements, so $s(n)$ counts the onto maps from M to N.
3. Assume $m{=}|M|{\geq}1$ for the stated Step 1 recurrence. If $m{=}0$, there is one onto map when $n{=}0$ and none when $n{>}0$; that case is separate from this recurrence.

$$f \text{ is onto} \Leftrightarrow |f(M)| = |N| = n$$

Challenge 4.A: Surjective maps

4.A Step 1 · Derive the recurrence for $s(k)$

1. There are k^m total maps $M{\to}A$. If the image has size j, first choose its j-element image B in $\mathrm{C}(k,j)$ ways, then choose a map onto B in $s(j)$ ways.
2. Summing over image sizes gives $k^m{=}\sum_{j=1}^{k}\mathrm{C}(k,j)s(j)$. There is no empty image because $m{\geq}1$.
3. The $j{=}k$ term is $s(k)$, since $\mathrm{C}(k,k){=}1$. Rearranging gives $s(k){=}k^m{-}\sum_{j=1}^{k-1}\mathrm{C}(k,j)s(j)$.

$$k^m = \sum_{j=1}^{k} \mathrm{C}(k,\, j)s(j)$$

$$s(k) = k^m - \sum_{j=1}^{k-1} \mathrm{C}(k,\, j)s(j)$$

Challenge 4.A: Surjective maps

4.A Step 2 · Check the proposed formula for $k{=}1$, 2, and 3

1. For $k{=}1$, $s(1){=}1^m{=}1$. The proposed sum is $\mathrm{C}(1,1)1^m{=}1$.
2. For $k{=}2$, $s(2){=}2^m{-}2s(1){=}2^m{-}2$. The guess gives $\mathrm{C}(2,1)(-1)^1 1^m{+}\mathrm{C}(2,2)2^m{=}2^m{-}2$.
3. For $k{=}3$, $s(3){=}3^m{-}3s(1){-}3s(2){=}3^m{-}3{\cdot}2^m{+}3$. The guess gives the same alternating expression. These cases support, but do not prove, $s(k){=}\sum_{j=1}^{k}\mathrm{C}(k,j)(-1)^{k-j}j^m$.

$$s(1) = 1$$

$$s(2) = 2^m - 2$$

$$s(3) = 3^m - 3 \cdot 2^m + 3$$

Challenge 4.A: Surjective maps

4.A Step 3a · Reverse the order of a triangular sum

1. The left-hand iterated sum first chooses j=1,…,k, then i=1,…,j. Its index pairs are exactly $\{(i,j):1\le i\le j\le k\}$.
2. If instead i is fixed first, i ranges from 1 to k and j ranges from i to k. Thus $\sum_{j=1}^{k}\sum_{i=1}^{j}(\dots) = \sum_{i=1}^{k}\sum_{j=i}^{k}(\dots)$.
3. Both sides add the same term once for every pair in that finite triangular region.

$$\sum_{j=1}^{k}\sum_{i=1}^{j}(\dots) = \sum_{i=1}^{k}\sum_{j=i}^{k}(\dots)$$

Challenge 4.A: Surjective maps

4.A Step 3b · Prove the binomial coefficient identity

1. For $0\le i\le j\le k$, $\mathrm{C}(k,j)\mathrm{C}(j,i)=k!/[j!(k-j)!]\cdot j!/[i!(j-i)!]$.
2. Cancel $j!$ to get $k!/[i!(j-i)!(k-j)!]$.
3. On the other hand, $\mathrm{C}(k,i)\mathrm{C}(k-i,j-i)=k!/[i!(k-i)!]\cdot(k-i)!/[(j-i)!(k-j)!]$, the same expression.

$$\mathrm{C}(k,\,j)\mathrm{C}(j,\,i) \;=\; \mathrm{C}(k,\,i)\mathrm{C}(k-i,\,j-i)$$

Challenge 4.A: Surjective maps

4.A Step 3c · Establish the convolution identity

1. Start with $\sum_{j=1}^{k}\mathrm{C}(k,j)g(j)$ and insert $g(j)=\sum_{i=1}^{j}\mathrm{C}(j,i)(-1)^{j-i}\,i^{m}$.
2. Use Step 3a to reverse the order, then Step 3b: $\mathrm{C}(k,j)\mathrm{C}(j,i)=\mathrm{C}(k,i)\mathrm{C}(k-i,j-i)$.
3. The result is $\sum_{i=1}^{k}\mathrm{C}(k,i)i^{m}\sum_{j=i}^{k}\mathrm{C}(k-i,j-i)(-1)^{j-i}$. Put $r=j-i$. The inner sum is $\sum_{r=0}^{k-i}\mathrm{C}(k-i,r)(-1)^{r}=(1-1)^{k-i}$.
4. It is zero unless $i=k$, when it equals 1. The whole expression therefore reduces to k^{m}, proving $k^{m}=\sum_{j=1}^{k}\mathrm{C}(k,j)g(j)$.

$$k^{m} \;=\; \sum_{j=1}^{k}\mathrm{C}(k,\,j)g(j)$$

Challenge 4.A: Surjective maps

4.A Step 4 · Prove $s(k)=g(k)$ by induction

1. For $k=1$, $s(1)=1$ and $g(1)=\mathrm{C}(1,1)(-1)^0 1^m=1$.
2. Assume $s(j)=g(j)$ for every $1\le j<k$. The recurrence gives $s(k)=k^m-\sum_{j=1}^{k-1}\mathrm{C}(k,j)s(j)$.
3. The convolution identity says $k^m=\sum_{j=1}^{k}\mathrm{C}(k,j)g(j)$, so isolating its final term gives $g(k)=k^m-\sum_{j=1}^{k-1}\mathrm{C}(k,j)g(j)$.
4. The induction hypothesis makes these right sides equal, hence $s(k)=g(k)$. This proves the formula for every $k\ge 1$.

$$s(k) \;=\; g(k) \;=\; \sum_{j=1}^{k} \mathrm{C}(k,\,j)(-1)^{k-j} j^m$$

Made by Thanate Amornjiraporn (

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